Frequency Distribution of GDP Groups

The FREQ Procedure

GDP_Group	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Above Average	87	55.06	87	55.06
Below Average	71	44.94	158	100.00

Descriptive Statistics of Happiness Score by GDP Group

The MEANS Procedure

Analysis Variable : Happiness_Score							
GDP_Group	N Obs	N	Mean	Median	Std Dev	Minimum	Maximum
Above Average	87	87	6.0497	5.9840	0.9353	3.8960	7.5870
Below Average	71	71	4.5499	4.5180	0.7807	2.8390	6.1300



Normality Assessment of Happiness Score by GDP Group

The UNIVARIATE Procedure
Variable: Happiness_Score
GDP_Group = Above Average

Moments			
N	87	Sum Weights	87
Mean	6.04967816	Sum Observations	526.322
Std Deviation	0.93534909	Variance	0.87487792

Moments			
Skewness	-0.1818486	Kurtosis	-0.8732175
Uncorrected SS	3259.31821	Corrected SS	75.239501
Coeff Variation	15.461138	Std Error Mean	0.10027995

Basic Statistical Measures			
Location		Variability	
Mean	6.049678	Std Deviation	0.93535
Median	5.984000	Variance	0.87488
Mode	5.192000	Range	3.69100
		Interquartile Range	1.64100

Tests for Location: Mu0=0				
Test	Statistic		p Value	
Student's t	t	60.32789	Pr > |t|	<.0001
Sign	M	43.5	Pr >= |M|	<.0001
Signed Rank	S	1914	Pr >= |S|	<.0001

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.970032	Pr < W	0.0409
Kolmogorov-Smirnov	D	0.07184	Pr > D	>0.1500
Cramer-von Mises	W-Sq	0.091517	Pr > W-Sq	0.1460
Anderson-Darling	A-Sq	0.643732	Pr > A-Sq	0.0924

Quantiles (Definition 5)	
Level	Quantile
100% Max	7.587
99%	7.587
95%	7.427
90%	7.286
75% Q3	6.853
50% Median	5.984
25% Q1	5.212
10%	4.800
5%	4.642
1%	3.896
0% Min	3.896

Extreme Observations			
Lowest		Highest	
Value	Obs	Value	Obs
3.896	48	7.427	25
4.194	42	7.522	106
4.218	20	7.527	38
4.332	18	7.561	59
4.642	128	7.587	136

Normality Assessment of Happiness Score by GDP Group

The UNIVARIATE Procedure
Variable: Happiness_Score
GDP_Group = Below Average

Moments			
N	71	Sum Weights	71
Mean	4.54991549	Sum Observations	323.044
Std Deviation	0.7806526	Variance	0.60941848
Skewness	0.06718428	Kurtosis	-0.3706555
Uncorrected SS	1512.48219	Corrected SS	42.6592935
Coeff Variation	17.1575186	Std Error Mean	0.09264642

Basic Statistical Measures			
Location		Variability	
Mean	4.549915	Std Deviation	0.78065
Median	4.518000	Variance	0.60942
Mode	.	Range	3.29100
		Interquartile Range	1.10100

Tests for Location: Mu0=0				
Test	Statistic		p Value	
Student's t	t	49.11054	Pr > |t|	<.0001
Sign	M	35.5	Pr >= |M|	<.0001
Signed Rank	S	1278	Pr >= |S|	<.0001

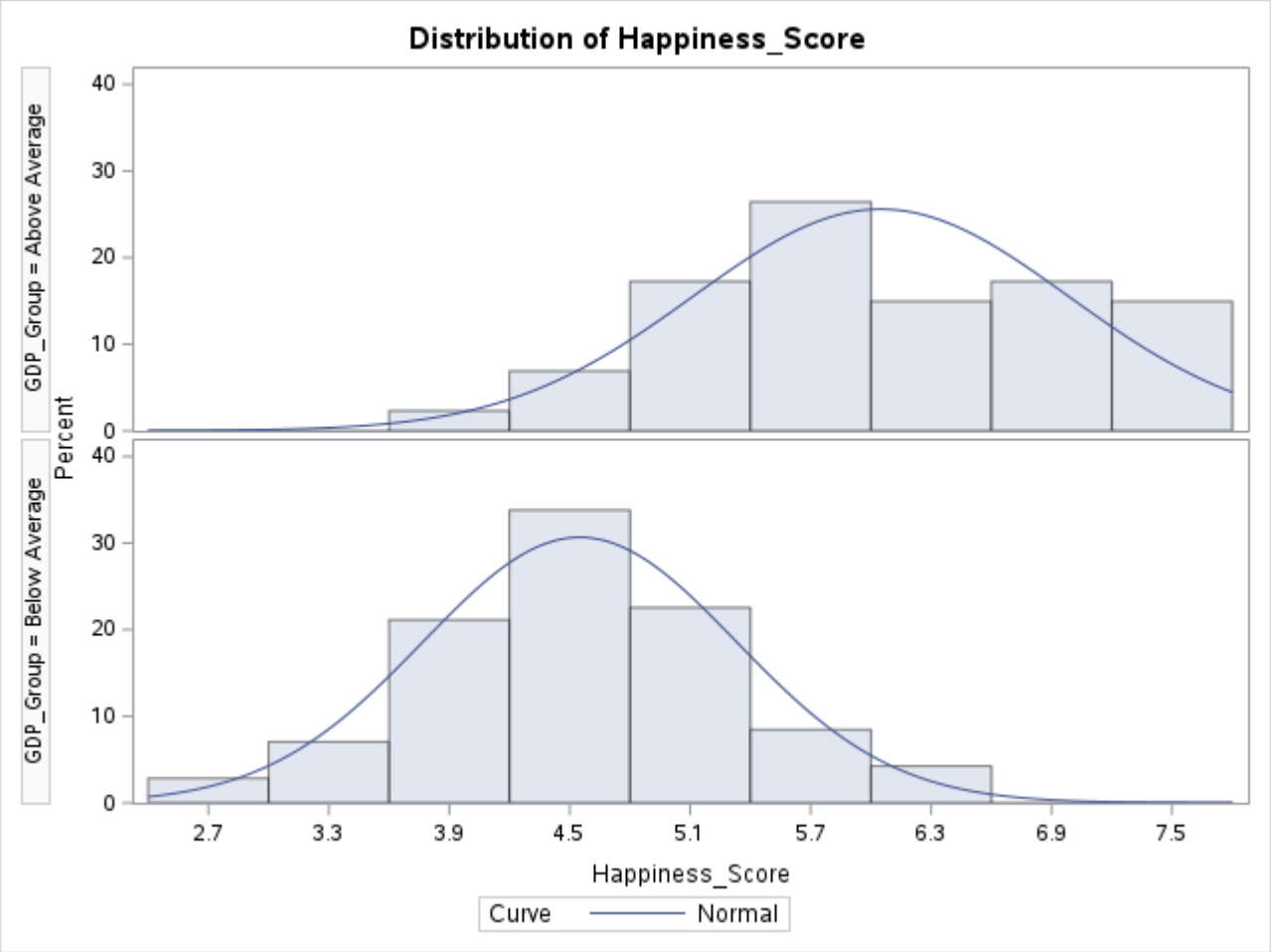
Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.984281	Pr < W	0.5186
Kolmogorov-Smirnov	D	0.047797	Pr > D	>0.1500
Cramer-von Mises	W-Sq	0.029945	Pr > W-Sq	>0.2500
Anderson-Darling	A-Sq	0.251926	Pr > A-Sq	>0.2500

Quantiles (Definition 5)	
Level	Quantile
100% Max	6.130
99%	6.130
95%	5.890
90%	5.709
75% Q3	5.057
50% Median	4.518
25% Q1	3.956
10%	3.655
5%	3.340
1%	2.839
0% Min	2.839

Extreme Observations			
Lowest		Highest	
Value	Obs	Value	Obs
2.839	142	5.889	93
2.905	22	5.890	16
3.006	137	6.003	153
3.340	14	6.123	53
3.465	119	6.130	43

Normality Assessment of Happiness Score by GDP Group

The UNIVARIATE Procedure



Normality Assessment of Happiness Score by GDP Group

The UNIVARIATE Procedure
GDP_Group = Above Average
Fitted Normal Distribution for Happiness_Score

Parameters for Normal Distribution		
Parameter	Symbol	Estimate
Mean	Mu	6.049678
Std Dev	Sigma	0.935349

Goodness-of-Fit Tests for Normal Distribution				
Test	Statistic		p Value	
Kolmogorov-Smirnov	D	0.07183992	Pr > D	>0.150
Cramer-von Mises	W-Sq	0.09151696	Pr > W-Sq	0.146
Anderson-Darling	A-Sq	0.64373193	Pr > A-Sq	0.092

Quantiles for Normal Distribution		
	Quantile	
Percent	Observed	Estimated
1.0	3.89600	3.87373
5.0	4.64200	4.51117
10.0	4.80000	4.85098
25.0	5.21200	5.41879
50.0	5.98400	6.04968
75.0	6.85300	6.68056
90.0	7.28600	7.24838
95.0	7.42700	7.58819

Quantiles for Normal Distribution		
	Quantile	
Percent	Observed	Estimated
99.0	7.58700	8.22563

Normality Assessment of Happiness Score by GDP Group

The UNIVARIATE Procedure
GDP_Group = Below Average
Fitted Normal Distribution for Happiness_Score

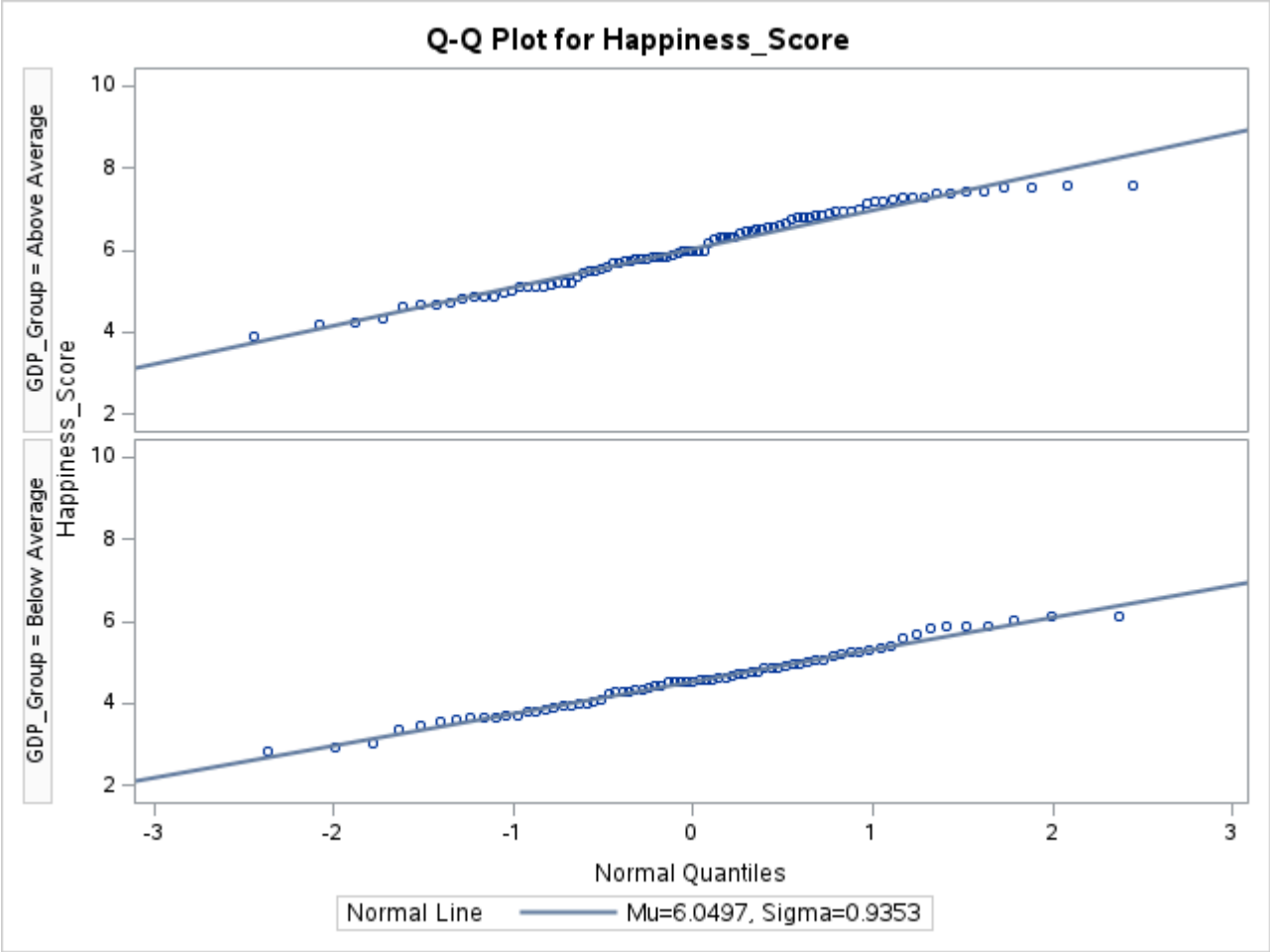
Parameters for Normal Distribution		
Parameter	Symbol	Estimate
Mean	Mu	4.549915
Std Dev	Sigma	0.780653

Goodness-of-Fit Tests for Normal Distribution				
Test	Statistic		p Value	
Kolmogorov-Smirnov	D	0.04779722	Pr > D	>0.150
Cramer-von Mises	W-Sq	0.02994517	Pr > W-Sq	>0.250
Anderson-Darling	A-Sq	0.25192620	Pr > A-Sq	>0.250

Quantiles for Normal Distribution		
	Quantile	
Percent	Observed	Estimated
1.0	2.83900	2.73385
5.0	3.34000	3.26586
10.0	3.65500	3.54947
25.0	3.95600	4.02337
50.0	4.51800	4.54992
75.0	5.05700	5.07646
90.0	5.70900	5.55036
95.0	5.89000	5.83397
99.0	6.13000	6.36599

Normality Assessment of Happiness Score by GDP Group

The UNIVARIATE Procedure



Independent Samples T-Test for Happiness Score by GDP Group

The TTEST Procedure

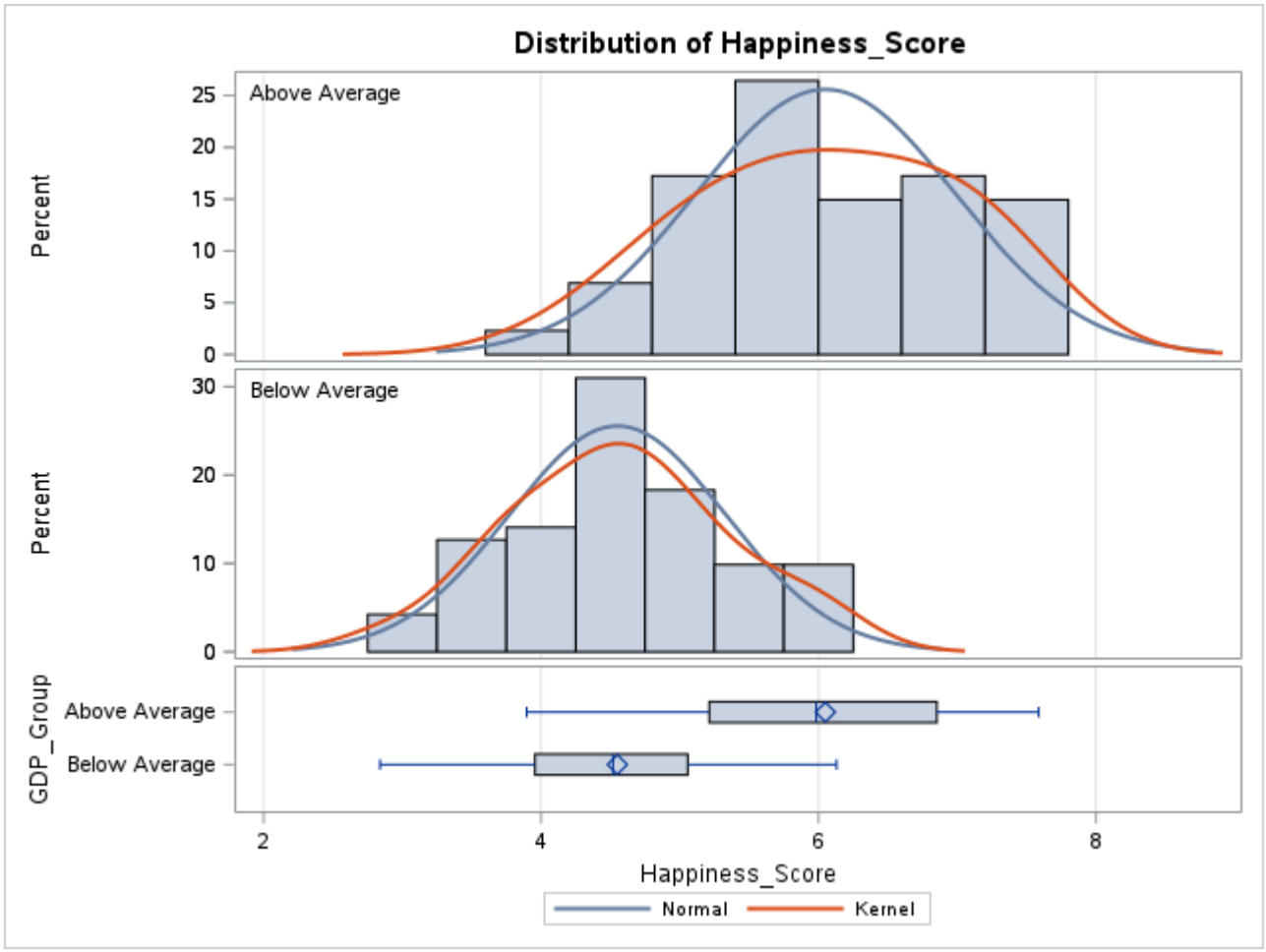
Variable: Happiness_Score

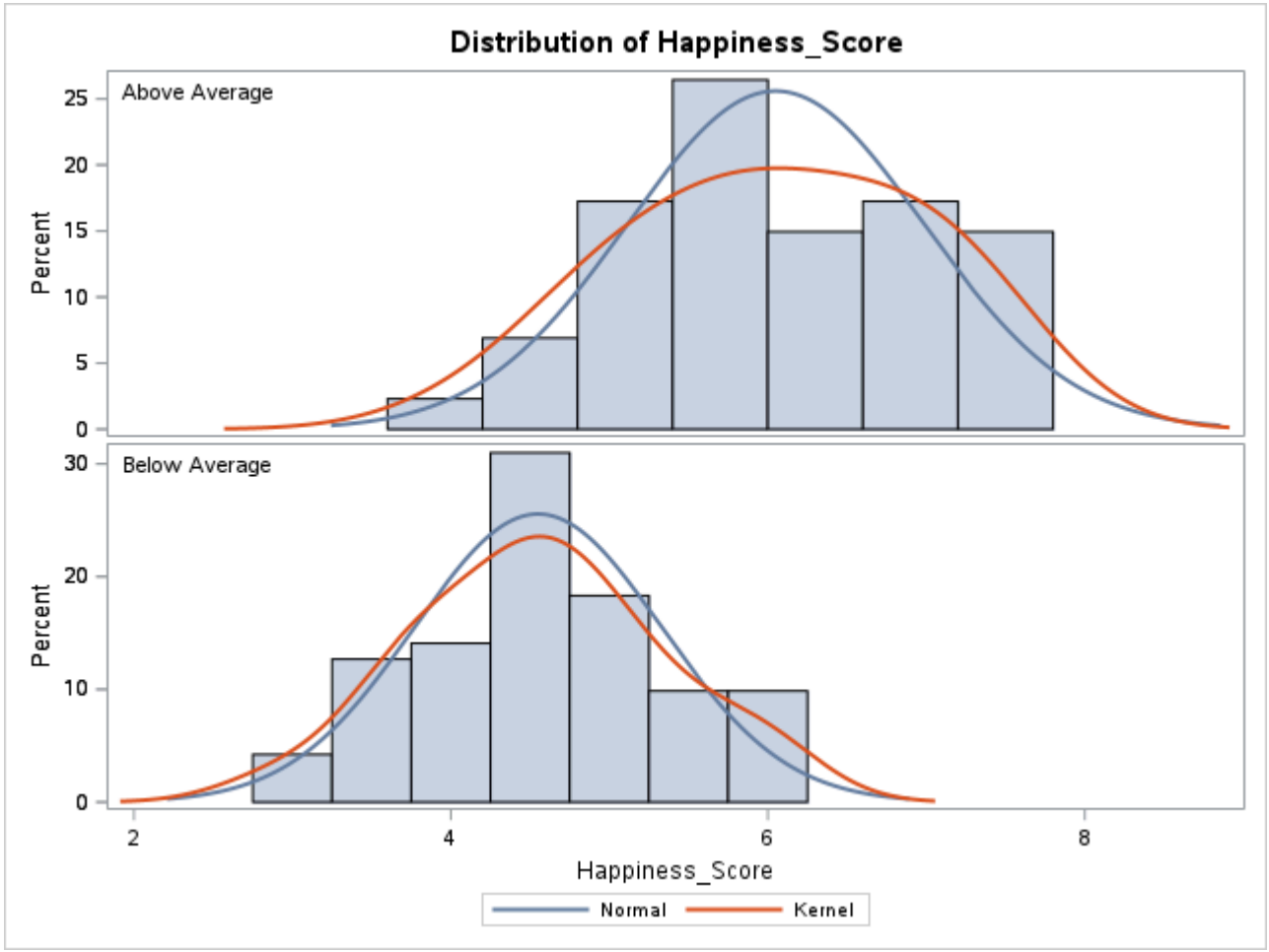
GDP_Group	Method	N	Mean	Std Dev	Std Err	Minimum	Maximum
Above Average		87	6.0497	0.9353	0.1003	3.8960	7.5870
Below Average		71	4.5499	0.7807	0.0926	2.8390	6.1300
Diff (1-2)	Pooled		1.4998	0.8693	0.1390		
Diff (1-2)	Satterthwaite		1.4998		0.1365		

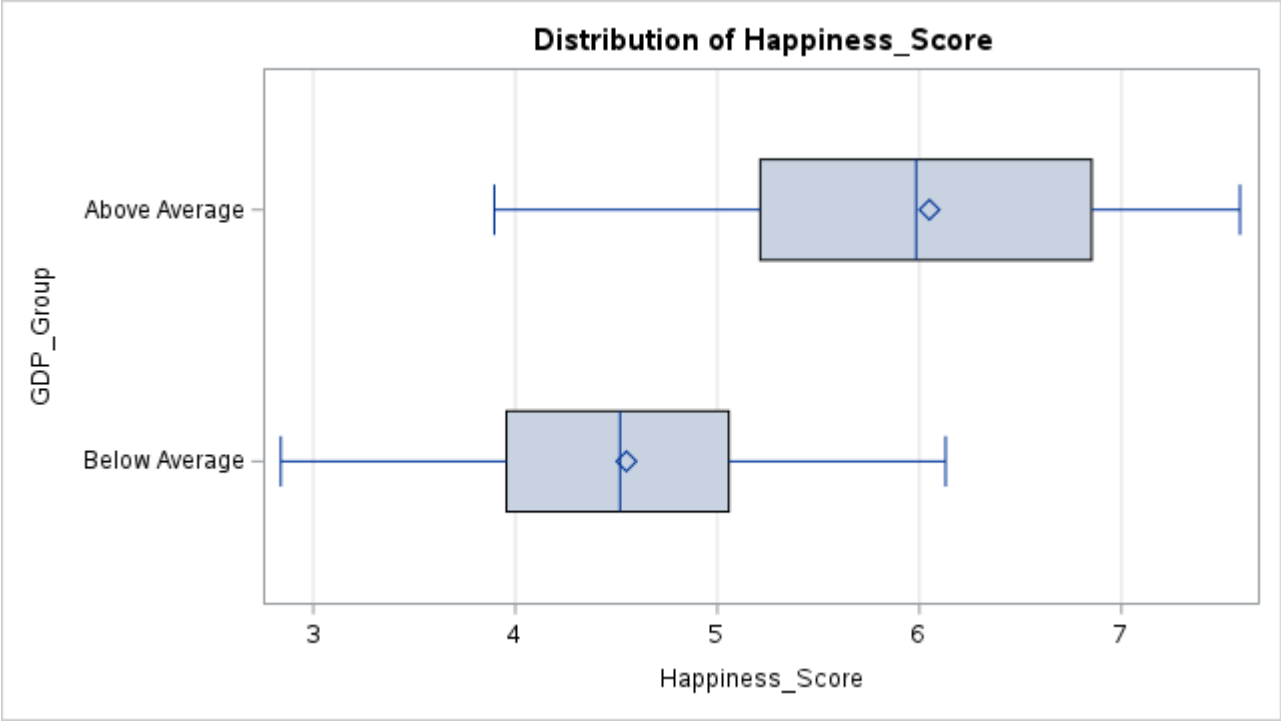
GDP_Group	Method	Mean	95% CL Mean		Std Dev	95% CL Std Dev	
Above Average		6.0497	5.8503	6.2490	0.9353	0.8140	1.0995
Below Average		4.5499	4.3651	4.7347	0.7807	0.6700	0.9354
Diff (1-2)	Pooled	1.4998	1.2251	1.7744	0.8693	0.7827	0.9778
Diff (1-2)	Satterthwaite	1.4998	1.2301	1.7694			

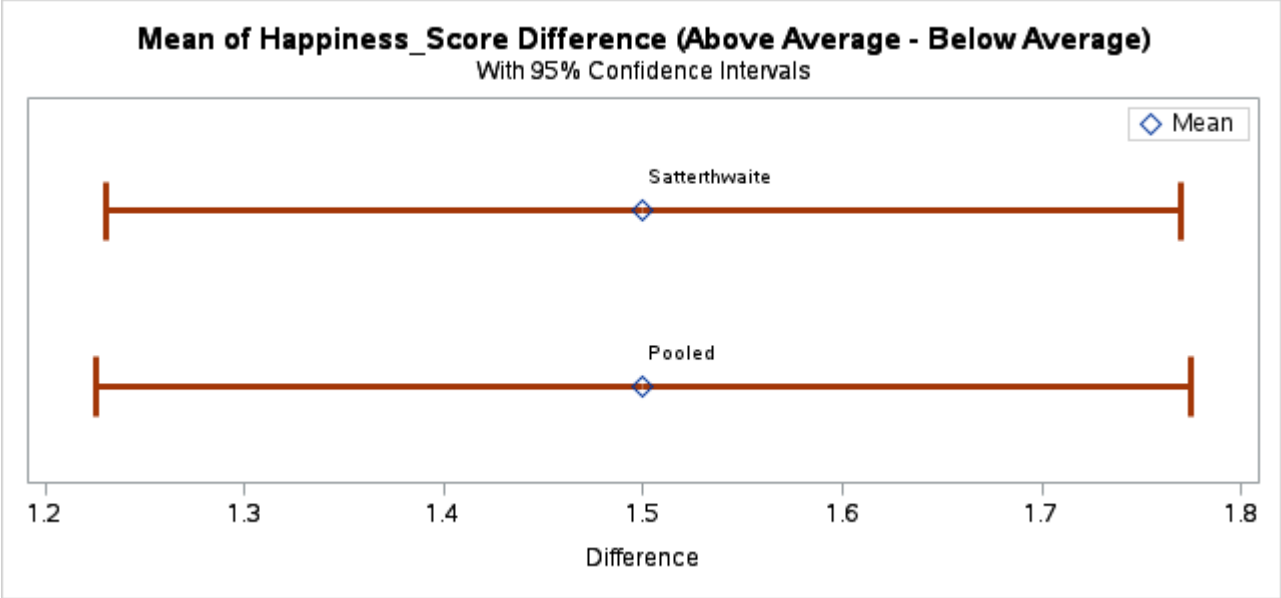
Method	Variances	DF	t Value	Pr > |t|
Pooled	Equal	156	10.79	<.0001
Satterthwaite	Unequal	155.91	10.99	<.0001

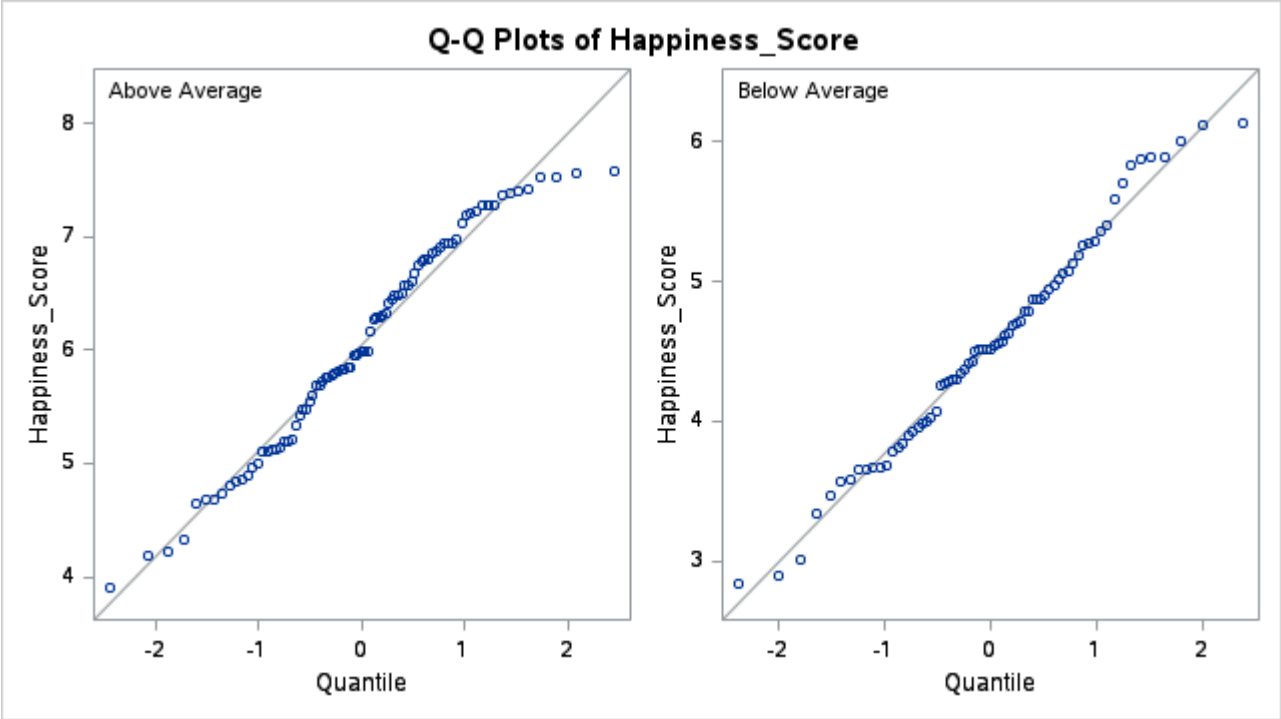
Equality of Variances				
Method	Num DF	Den DF	F Value	Pr > F
Folded F	86	70	1.44	0.1184


Distribution of Happiness_Score
Above Average
Below Average
Percent
GDP_Group
Above Average
Below Average
Happiness_Score
Normal
Kernel


Distribution of Happiness_Score
Above Average
Below Average
Percent
Happiness_Score
Normal
Kernel


Distribution of Happiness_Score
GDP_Group
Above Average
Below Average
3
4
5
6
7
Happiness_Score


Mean of Happiness_Score Difference (Above Average - Below Average)
With 95% Confidence Intervals
Mean
Satterthwaite
Pooled
1.2
1.3
1.4
1.5
1.6
1.7
1.8
Difference


Q-Q Plots of Happiness_Score
Above Average
Below Average
Happiness_Score
Quantile
8
7
6
5
4
3
-2
-1
0
1
2